

Joel E. Castro Hernandez

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EDUCATION

- University of Southern California, Viterbi School of Engineering** Aug 2026 – May 2028
M.S. Computer Science
Coursework: Deep Learning (CSCI 566), Algorithms (CSCI 570), Computer Vision (CSCI 677)
- University of California, Berkeley** Jul 2022 – May 2026
B.A. Computer Science, SEED Scholar Honors Program, GPA: 3.5
Relevant Coursework: Computer Vision & Computational Photography (Profs. Efros & Kanazawa), Computer Graphics & Imaging, Intro to Machine Learning, Signals & Systems, 3D Modeling & Animation, Data Structures & Algorithms

RESEARCH EXPERIENCE

- Stanford Robotics and Embodied AI Lab (REAL) [Prof. Shuran Song]** Mar 2025 – Aug 2025
- Co-authored a CoRL conference publication (in-review) with Prof. Shuran Song and Stanford graduate students [1]
 - Trained and deployed diffusion-based generative policies as baselines against our novel policy architecture, iterating on sampling schedules and evaluation protocol [1]
 - Designed a novel robotic handwriting task (3D-printed pen gripper) for handheld data collection (450+ demos), transferred to ARX5 robot; built a Grounded-SAM2/OpenCV pipeline for evaluation [1]
- UCSF Cardiac Vision Laboratory [Prof. Jan Christoph]** Sep 2024 – Jan 2025
- Worked with state-of-the-art differentiable rendering and 3D/4D scene reconstruction methods to optimize novel-view visualization of cardiac optical and ultrasound data
 - Developed a multi-view camera calibration and visualization tool, automating the calibration process, eliminating repeated manual execution, and minimizing downtime by 90%
- CMU Software and Societal Systems Dept. [Prof. Jonathan Aldrich]** May 2024 – Aug 2024
- First-authored SPLASH publication on a type theory for enhancing diagramming tools, categorically coding 150+ open-source visualization tools' source code (e.g., AMD GPUOpen, Torchview) for statistically significant results [2]
- Berkeley Staller Labs [Prof. Max Staller]** Aug 2022 – May 2024
- Explored SHAP analysis on an in-lab CNN to increase interpretability, determining which features it weighs most when predicting protein activation domains [3]
 - Streamlined all-atom Monte Carlo simulation pipeline on a computer cluster with Bash scripting, cutting runtime over 85% and documenting the workflow to train 2 peers

PUBLICATIONS

- A. Patel, B. Pekarek, **J. E. C. Hernandez**, and S. Song, "Behavior prompting policy: Demonstrations as prompts for manipulation," *arXiv preprint arXiv:2606.30457*, 2026 [Webpage] [Paper]
- J. E. Castro Hernandez** and O. G. John, "Understanding program visualizations in the wild," in *Companion Proceedings of the 2024 ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications: Software for Humanity (SPLASH Companion '24)*, pp. 49–51, 2024 [Webpage] [Paper]
- C. LeBlanc, P. Agarwal, J. Demaray, G. Hu, M. Zintel, A. Lam, **J. E. Castro Hernandez**, and M. Staller, "Interpretable biophysical neural networks of transcriptional activation domains separate roles of protein abundance and coactivator binding," *Cell Systems*, 2026 [Paper]

PROJECTS

- Neural Radiance Fields (NeRF) | CS280, UC Berkeley | Webpage** Spring 2026
- Built an end-to-end multi-view 3D reconstruction pipeline: calibrated smartphone camera intrinsics via OpenCV ArUco detection, solved camera poses, and constructed an undistorted, pose-registered dataset
 - Implemented a Neural Radiance Field from scratch in PyTorch (positional encoding, MLP-based volumetric radiance field, differentiable ray marching), sweeping hyperparameters and PSNR novel-view synthesis evaluation
- Diffusion Models & Flow Matching from Scratch | CS280, UC Berkeley | Webpage** Spring 2026
- Implemented diffusion sampling loops (DeepFloyd IF) for generation, inpainting, and illusion synthesis
 - Built a UNet denoiser, then a flow-matching model on MNIST, comparing iterative vs. one-step sampling
- Minecraft Fabric Mod – CS184 Final Project Honorable Mention | GitHub** Jan 2025 – May 2025
- Implemented procedural sphere generation, texture mapping, and quaternion-based rotation mechanics for physically accurate interactions
 - Contributed to shader enhancements, video production, and final deliverables, helping the project earn Honorable Mention in CS184, and receiving over 1.1K downloads on CurseForge from the Minecraft community

TEACHING EXPERIENCE

- Computer Graphics (CS184) TA [1 semester]** - UC Berkeley Dec 2025 – May 2026
- Performed weekly office hours and graded technical homework and exams on topics such as ray tracing, physical simulation, and rendering
- Undergraduate Graphics Group (UCBUGG) Facilitator [3 semesters]** - UC Berkeley Jan 2025 – May 2026
- Presented weekly lectures and mentored student groups through end-to-end production of animated shorts
- Computer Architecture (CS61C) Tutor/Grader [1 semester]** - UC Berkeley Aug 2024 – Dec 2024
- Addressed an average of 30+ student questions weekly during office hours on C, RISC-V, and Logisim concepts

PRESENTATIONS

- Stanford SURF Symposium 2025:** Lightning talk and poster presentation of Behavior Prompting Policy.
- SPLASH 2024** (Pasadena, CA): Poster and oral presentation of *Understanding Program Visualizations in the Wild*.
- SACNAS 2024** (Phoenix, AZ): Poster presentation of *Understanding Program Visualizations in the Wild*.

ACADEMIC SERVICE

- (Berkeley) Anova – On-site Tutor; Publicity/Curriculum Committee** Jan 2023 – May 2026
- Taught coding concepts in Python and Scratch to high school students weekly, enhancing their understanding of CS
 - Developed Arduino curricula to improve tech accessibility for students in under-resourced communities

AWARDS

- **Jack Kent Cooke Graduate Scholarship** - Highly competitive national scholarship for graduate studies 2026
- **CODE2040 Fellowship** - Fellowship for top Black and Latinx computer science students 2024
- **eBay Pathways Fellowship** - Fellowship supporting underrepresented students in tech 2024
- **Jack Kent Cooke Scholarship** - Highly competitive national scholarship for undergraduate studies 2022
- **Cal Alumni Leadership Award** - Merit-based scholarship recognizing leadership at UC Berkeley 2022
- **SEED Scholars Honors Program** - STEM honors program for academic excellence and research 2022

SKILLS

Languages: Python, C/C++, Java, Bash, SQL, JavaScript/HTML/CSS

ML & Graphics: PyTorch, NeRF, 3D Gaussian Splatting, Diffusion & Flow-Matching Models, Differentiable Rendering, CUDA

Software & Tools: ROS, Unreal Engine, Git, Arduino, Logisim

Design: Autodesk (Maya, Fusion, Inventor), Adobe (Photoshop, Premiere Pro, After Effects), Nomad Sculpt

Spoken Languages: English & Spanish (Native/Bilingual)